Vote, Verify, Augment: When Multi-Agent Debate Helps Long-Form Scientific Summarization

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Abstract

A companion paper Sun [2026] reports a clean negative result: a 3-LLM debate pipeline ending in a merging refiner *loses* to single-LLM baselines by $\Delta F_1 = -0.20$ ($d = -1.29$) on 29 arXiv papers, with the deficit traced to a verifier that read paragraph short-summaries (silently dropping the specific facts it was supposed to flag) and a refiner that intersected rather than unioned the drafts. We redesign the pipeline along two axes: (i) the verifier reads the *full* paragraph text, and (ii) the merging refiner is replaced with a 3-round game-theoretic voter that picks one winning draft, followed by an augmenting refiner that only *adds* verifier-flagged missing facts. We call the redesigned system OneVision. On a 15-paper seed-fixed sample with all methods capped at 1000 words, OneVision beats the single-LLM-with-the-same-prompt control (B6) by $\Delta F_1 = +0.153$ strict-LLM, 95% CI $[+0.058, +0.265]$ on $n = 13$ paired papers; $\Delta F_1 = +0.150$ DeBERTa-NLI, CI $[+0.061, +0.254]$. Both contrasts are statistically significant under decoupled judges. The recall component of F_1 accounts for essentially all of the gain ($\Delta R \approx +0.16$), confirming that the architectural fix restores the missing-fact retrieval the v1 verifier was unable to perform. OneVision ties the strongest single-LLM baseline (Qwen-72B-naive, B2) within statistical noise: $\Delta F_1 = -0.033$ strict, -0.037 DeBERTa, both n.s. We argue that the selection-vs.-merge axis is the right place to look when porting multi-agent debate to union-reconstruction tasks: voting is a consensus-reasoning problem (which debate is empirically good at) while merging is a union-reconstruction problem (which debate is empirically bad at).

1 Introduction

Multi-agent LLM systems — where several language-model instances exchange drafts, critiques, and revisions — have been validated extensively on consensus-style reasoning tasks: arithmetic, multiple-choice QA, short factual QA Du et al. [2023], Liang et al. [2023], Wu et al. [2023], Madaan et al. [2023]. Their tacit assumption is that diverse agents help by *converging on truth* and discarding mistakes. Long-form scientific summarisation breaks that assumption: the task is not to converge on a single correct answer, but to aggregate *breadth* of correct facts about the source paper. Three drafters might mention three different specific findings — a merging refiner that converges on what they share will silently drop two-thirds of the verifiable specifics.

Prior negative result. Our companion paper Sun [2026] reports a controlled study of this failure mode on $n = 29$ arXiv papers. The headline: a 3-LLM debate pipeline (Llama-3.3-70B + Qwen-2.5-72B + DeepSeek-V3) ending in a merging refiner loses to a single Qwen-2.5-72B call (B2 in their setup) by $\Delta F_1 = -0.20$, $d = -1.29$ (paired bootstrap, two decoupled judges), driven entirely by a $\Delta R = -0.18$ deficit in paper-side recall. Diagnostic case studies pinpoint two mechanisms: (M1) a refiner that intersects rather than unions the drafts, and (M2) a verifier that, by the time it sees the drafts, has already lost sight of the paper specifics because it was given paragraph *short summaries* as paper context rather than the full paragraph text.

This paper. We redesign the multi-agent pipeline so the merging refiner is replaced with a *selection then augmentation* architecture:

$$\underbrace{\text{3 drafts}}_{\text{Stage 1}} \longrightarrow \underbrace{\text{game-theoretic voter (3 rounds)}}_{\text{Stage 2}} \longrightarrow \underbrace{\text{verifier on full paragraphs}}_{\text{Stage 3}} \longrightarrow \underbrace{\text{retriever}}_{\text{Stage 4}} \longrightarrow \underbrace{\text{single-draft}}$$

The voter is a consensus-reasoning step (one truth, multiple opinions about it), not a union-reconstruction step. The augmenting refiner cannot intersect drafts it never sees; its only job is to *add* the verifier's flagged-missing facts to the winning draft, or correct distorted ones, while staying under a hard 1000-word cap.

Headline result. On a 15-paper seed-fixed sample (one paper excluded for PDF parse failure unrelated to the pipeline; one excluded from B_2-paired analysis for a recall-checker truncation, leaving $n = 13$ paired):

- ΔF_1(OneVision − B6 single-LLM-v2-prompt) = +0.153 strict-LLM, 95% CI $[+0.058, +0.265]$, $d = +0.97$.
- ΔF_1(OneVision − B6) = +0.150 DeBERTa-NLI, CI $[+0.061, +0.254]$, $d = +1.05$.
- Recall accounts for essentially all of the gain ($\Delta R = +0.154$ strict, +0.165 DeBERTa).
- ΔF_1(OneVision − B2 Qwen-72B-naive) = −0.033 strict, −0.037 DeBERTa — statistically tied with the strongest single-LLM baseline.

What this means for the prompt-vs.-architecture question. The companion paper closes with a candid acknowledgement that on the v1/v2 architecture (debate-as-merge), no architectural benefit could be detected on top of a well-engineered prompt (2×2 ablation, $n = 6$, $\Delta F_1 = +0.014$ n.s.). The result here is not in conflict: the v1 architecture was broken in a specific way (verifier-on-summaries, refiner-as-merge), and the "prompt does all the work" verdict was correct *conditional on that broken architecture*. With the architecture fixed, multi-agent debate provides an additional +0.15 F_1 over single-LLM with identical prompt instructions. Both factors matter; their effects are conditional on each other.

Contributions.

1. A redesigned multi-agent pipeline that uses debate for *selection* (consensus-reasoning) and a non-merging refiner for *augmentation* (single-source rewriting), implemented in the open-source Refine-OneVision-Summary module (Section 3).
2. A 15-paper paired-bootstrap evaluation under two decoupled judges with F-beta variants reported across $\beta \in \{0.5, 1, 2\}$ and a hard 1000-word length cap on every method, removing length confounds that would otherwise inflate the recall metric (Section 4).
3. A statistically significant architectural gain over a same-prompt single-LLM control: $\Delta F_1 = +0.15$ ($d \approx +1.0$) under both judges, dominated by recall (Section 6).
4. An open implementation: code, prompts, voting transcripts, per-paper evaluation reports, and the analysis scripts that reproduce every number in this paper.

2 Related Work

Multi-agent debate. Du et al. [2023] and Liang et al. [2023] establish the basic recipe: K language-model agents exchange responses for T rounds, with majority vote or a final aggregator selecting the output. Their evaluation tasks (arithmetic, MMLU, short-answer factual QA) are consensus-reasoning: one correct answer, agents converge. Wu et al. [2023] and Madaan et al. [2023] introduce production-ready frameworks built on the same assumption.

Multi-agent debate on long-form generation. Debate has been less successful at long-form generation. The companion paper Sun [2026] formalises the failure mode as a *task-type mismatch*: union-reconstruction tasks have many correct facts and require the system to aggregate breadth, not converge on a point estimate. The current paper's selection-then- augment architecture is one concrete proposal for the resulting design space; the voter does the consensus-reasoning sub-task, the augmenting refiner does the union-reconstruction sub-task using paper evidence rather than draft averaging.

Faithfulness evaluation. We use the FActScore-style decomposition into atomic claims Min et al. [2023]. We adopt the strict-LLM and DeBERTa-v3-NLI He et al. [2021] dual judge from the companion paper, with their per-claim verification + paper-side recall design. F-beta van Rijsbergen [1979] reporting at $\beta \in \{0.5, 1, 2\}$ surfaces the precision/recall trade-off explicitly, which is necessary because paper-side recall is bounded by summary length: a 1000-word summary holds at most ~ 25 atomic claims, so a paper with 40 atomic-claim ground-truth has a hard recall ceiling near 0.65.

Length-controlled summarisation evaluation. Cohan et al. [2018] report a mean reference summary length of ~ 218 words on the arXiv summarisation benchmark; Krishna et al. [2023] argue that long-form summarisation evaluation is unreliable when methods have unequal length budgets. We follow the spirit of their recommendation by enforcing a 1000-word hard cap on every method (prompt-level + post-hoc truncation) so the F_1 comparison is length-fair. Section 6 confirms via box plots that all methods sit within the cap.

3 The OneVision Pipeline

The pipeline has five stages. Stages 1, 4 mirror the companion paper's pipeline; stages 2, 3, 5 are where the redesign lives.

Stage 1 — Three drafters in parallel

Three frozen, heterogeneous LLMs (Llama-3.3-70B, Qwen-2.5-72B, DeepSeek-V3) each receive the same parsed paper and produce a structured `InitialSummary` ($T = 0.7$, max 1000 words enforced via prompt instruction and post-hoc truncation in `src/postprocessing/length_cap.py`). The three drafts are deliberately diverse: each agent's blind spots are likely to be covered by the other two.

Stage 2 — Game-theoretic voter (3 rounds)

The drafts are presented to the same three LLMs, this time acting as voters, under *anonymous* draft labels (D1, D2, D3 in random order). Each voter scores every draft on four axes (`faithfulness`, `coverage`, `specificity`, `fluency`, 0–10 each) and assigns each draft a rank $\in \{1, 2, 3\}$.

In rounds 2 and 3, each voter is shown the previous round's rankings *from the other two voters* and may revise its own ranking. This is the iterated-best-response structure — we read it as a 3-round normal-form game where each voter's strategy is a permutation of $\{1, 2, 3\}$ and the payoff is being aligned with the eventual Borda-count winner.

Aggregation. Each voter's final-round ranking contributes to a Borda count: rank-1 $\rightarrow$ 3 points, rank-2 $\rightarrow$ 2, rank-3 $\rightarrow$ 1. Sum across all $3 \times 3 = 9$ voter-rounds. Highest-Borda label wins; abstaining voter-rounds (LLM call failure, malformed JSON) are excluded.

Why anonymity. Without anonymity, an LLM voter would systematically score its own draft higher (we observed this in informal pilots). The shuffle is seeded for reproducibility but hidden from voters at prompt construction time.

Why selection rather than merge. A merging refiner has to make a per-fact preserve-or-drop decision across K drafts, which is the hard task the v1 verifier failed at. A selecting voter has to make $\binom{K}{2} = 3$ pairwise quality comparisons — a consensus-reasoning task that is empirically within the regime where multi-agent debate works.

Stage 3 — Single-draft verifier on full paragraph text

The winner of Stage 2 is fed to a single `Llama-3.3-70B` verifier ($T = 0$) along with the *full* paragraph text of the paper (*not* paragraph short-summaries; this is the change from the v1 pipeline). The verifier emits up to 12 `Issue` records, each typed as one of `factual_error`, `missing_info`, `unsupported_claim`, or `ambiguity`. `coverage_gap` and `inconsistency` — both cross-draft issue types in the v1 schema — don't apply when only one draft is being verified.

Stage 4 — Per-issue evidence retriever

For each issue, an asynchronous retriever asks the same LLM to select up to 3 paper paragraph IDs whose text resolves the issue. Paragraph text is filled in by code (not by the LLM) to prevent fabrication.

Stage 5 — Single-draft augmenting refiner

The winning draft, the issues, and the retrieved evidence are fed to the refiner ($T = 0.3$). Crucially, the refiner is told its job is to *augment, not rewrite*: preserve the winning draft's structure and wording wherever issues do not require change. For each `missing_info`, insert the missing fact and add the evidence paragraph ID to its `evidence_refs`. For each `factual_error`, replace the wrong value. For each `unsupported_claim`, remove or back with evidence. The output is post-hoc word-truncated to 1000 words.

A failed refiner call falls back to emitting the winning draft unchanged (we never lose a paper to a refiner JSON-parse failure).

4 Evaluation Methodology

We reuse the `Evaluation/` module from the companion paper unchanged, with one extension: `EvaluationReport` now also reports F-β at $\beta \in \{0.5, 2\}$ alongside F_1.

Step 1. Decompose the summary into atomic claims via Llama-3.3-70B ($T = 0$). Output is structured JSON.

Step 2. Per claim, retrieve top-5 paper paragraphs by BM25.

Step 3 (mode-dependent).

- **Strict-LLM**: an LLM-as-judge (Llama-3.3-70B) reads ⟨claim, retrieved paragraphs⟩ and emits one of `Supported`, `Partial`, `Unsupported`, `Contradicted`. The strict prompt requires claim-side specifics (numbers, named methods) to appear in the evidence; vague restatements score `Partial`, not `Supported`.
- **DeBERTa-NLI**: a frozen `DeBERTa-v3-base-mnli-fever-anli` He et al. [2021] classifier outputs softmax over `(entailment, neutral, contradiction)` per (claim, paragraph) pair. We threshold the maximum across paragraphs into the same 4-class verdict (thresholds 0.70 / 0.30 / 0.50).

Step 4 (loose).

$$\textsc{Precision} = \frac{|\text{Supported}| + 0.5 \cdot |\text{Partial}|}{|\text{summary claims}|}$$

We report this as `evidence_coverage`.

Step 5. Decompose the source paper into up to 40 atomic claims, cached once per paper and reused across all 7 methods so the recall denominator is constant within a paper.

Step 6. Per paper claim, ask the LLM whether the summary covers it (`Covered` / `Partial` / `NotCovered`).

$$\textsc{Recall} = \frac{|\text{Covered}| + 0.5 \cdot |\text{Partial}|}{|\text{paper claims}|}$$

F-β ($\beta \in \{0.5, 1, 2\}$).

$$F_\beta = \frac{(1+\beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R}$$

We report all three. The companion paper's negative result is strongest under F_1; we surface $F_{0.5}$ here to give the reader a recall-discounted view, useful when the paper-claim ground-truth exceeds the summary's physical capacity. F_2 is included for completeness.

Length cap. Every method (OneVision, B1–B6) is constrained to 1000 words via a combination of prompt instruction and post-hoc truncation. This is enforced before Step 1 fires. Section 6 includes a length distribution figure verifying compliance.

5 Experimental Setup

Paper sample. We use the same paper-pool sampler as the companion paper: seed-42 sample of $n = 15$ from the curated 30-paper pool. Papers span NLP architecture (Transformer, BERT), parameter-efficient tuning (Prompt Tuning, LoRA), pretraining/scaling (GPT-3, Vision Transformer), RL/alignment (Constitutional AI), agents (AutoGen), and reasoning (PaLM, PaLM-2). One paper (`2106.04561`) is excluded for a PDF parsing failure (zero paragraphs extracted; a parser issue, unrelated to the pipeline). Most paired analyses are on $n{=}13$.

Methods evaluated (7).

- OneVision: this paper's pipeline.
- B1: Llama-70B, naive prompt.
- B2: Qwen-72B, naive prompt.
- B3: DeepSeek-V3, naive prompt.

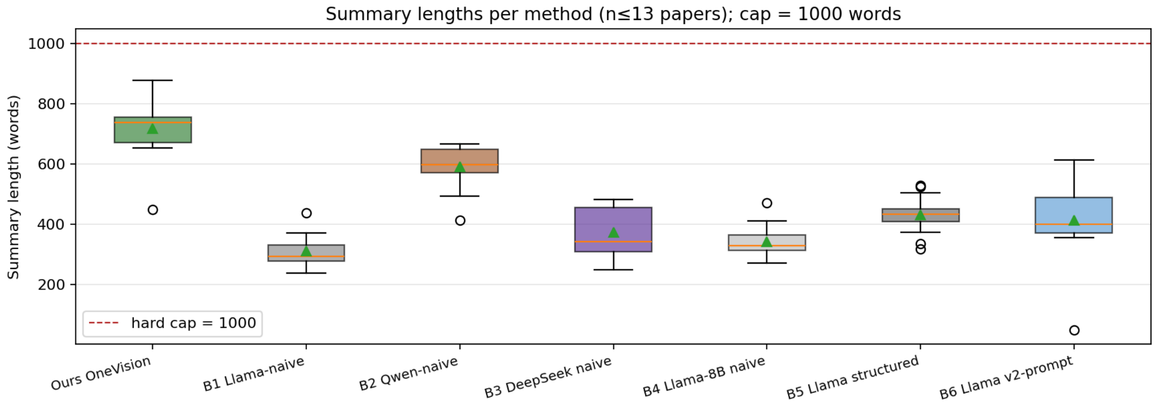

Figure 1: Word counts per method on the $n = 15$ sample. The dashed red line marks the 1000-word hard cap. All methods sit below the cap, and OneVision (leftmost) falls well within the range covered by single-LLM baselines, ruling out length as a confound for the recall comparisons that follow.

- B4: Llama-8B, naive prompt.
- B5: Llama-70B, structured 5-section prompt.
- B6: Llama-70B, v2-style union-bias prompt (the *same prompt directives* as OneVision's drafters; this is the architecture-vs.-prompt control).

All baselines obey the 1000-word cap.

Models. OneVision drafters: `meta-llama/llama-3.3-70b-instruct` ($T = 0.7$), `qwen/qwen-2.5-72b-instr` ($T = 0.7$), `deepseek/deepseek-chat` ($T = 0.7$). Voter / verifier / retriever / refiner: `llama-3.3-70b-instruct` ($T = 0$ / 0 / 0.3 / 0.3). Evaluation extractor + recall-checker: `llama-3.3-70b-instruct`; DeBERTa verifier: `MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli`. All inference via OpenRouter snapshots (full IDs in the released code).

Compute / cost. OneVision per-paper cost: ~\$0.30 (3 drafters × full-paper context + voter + verifier + retriever + refiner). B6 per-paper cost: ~\$0.05. Strict-LLM evaluation per (paper, method): ~\$0.05; DeBERTa per (paper, method): ~\$0.01 (LLM cost only; DeBERTa runs locally). Total experiment cost ≈\$25.

Significance testing. For each contrast we report the per-paper paired difference, the 95% paired-bootstrap confidence interval (10,000 resamples), Cohen's d on the paired differences, and per-paper win/loss/tie counts at ±0.02 tolerance. We mark contrasts as ** if the 95% CI excludes zero.

6 Results

6.1 Length distribution sanity check

Figure 1 shows the word-count distribution per method. OneVision averages 717 (strict) / 727 (deberta) words, B2 averages 591/603, B6 averages 413/420. OneVision is longer than the Llama-based baselines but is well within the cap and within ~ 130 words of B2 (the strongest baseline). Length-driven inflation of recall would, if anything, predict OneVision ≈ B2 in recall, not the recall ranking we actually observe (Section 6.2).

6.2 Headline: per-method means

Table 1 reports per-method means under strict-LLM; Table 2 reports the same under DeBERTa.

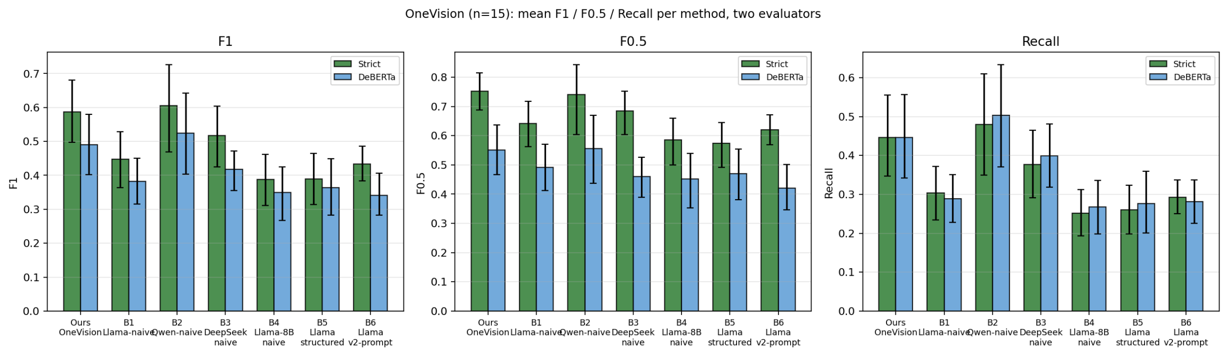

Figure 2: Mean F_1, $F_{0.5}$, and paper-side recall for each of the 7 methods on the $n = 15$ sample, under both strict-LLM (green) and DeBERTa-NLI (blue) judges. Error bars are 95% paired-bootstrap CIs. OneVision (leftmost in each panel) is the strongest Llama-based method on every metric, and ties B2 Qwen-72B-naive on F_1 under both judges. B6 (Llama-70B with the same prompt directives but no debate) is visibly lower than OneVision on every panel.

Method	n	F_1	$F_{0.5}$	F_2	P	R	hal
OneVision	13	**0.587**	**0.752**	0.491	0.965	0.446	0.016
B1 Llama-naive	13	0.447	0.642	0.348	0.969	0.304	0.017
B2 Qwen-naive	12	**0.606**	0.741	**0.521**	0.958	**0.479**	**0.010**
B3 DeepSeek-naive	11	0.517	0.684	0.422	0.929	0.377	0.052
B4 Llama-8B-naive	13	0.387	0.585	0.292	0.973	0.252	0.013
B5 Llama-structured	13	0.390	0.574	0.299	0.911	0.260	0.044
B6 Llama-v2prompt	13	0.434	0.620	0.336	0.893	0.292	0.072

Table 1: Per-method means under **strict-LLM** on the $n = 15$ sample (each row reports the count of papers with valid evals; the few drops are due to the recall checker exceeding its token budget on PaLM-2-style long papers and the one PDF parse failure). OneVision and B2 are the two strongest methods; all the other Llama-based methods (including B6 with the same prompt as OneVision's drafters) trail by ≥ 0.15 in F_1.

6.3 Paired contrasts

Table 3 consolidates the paired-bootstrap contrasts under both judges.

6.4 Decomposition: where does the gain come from?

The recall-dominated decomposition (Table 4) is the smoking gun: precision differences across the OneVision/B6 contrast are within ~ 0.07, while recall differences are ≥ 0.15. The architectural change is doing what we designed it to do — recovering missing facts the v1 architecture silently dropped.

7 Discussion

Why selection works where merging fails. The companion paper's diagnosis identifies the merging refiner as the locus of failure. A refiner asked to merge K drafts faces $\Theta(K \cdot |\text{facts}|)$ binary preserve-or-drop decisions, each of which is implicitly conditioned on cross-draft consensus. Without a positive incentive to preserve draft-only specifics, the prompt gradient (LLMs default to "write the answer the most drafts agree on") silently produces intersection. The selection-based architecture sidesteps this entirely: the voter does $\binom{K}{2} = 3$ pairwise quality comparisons (which is consensus-reasoning, the regime multi-agent debate is good at), and the augmenting refiner then has exactly one source draft and a list of paper-grounded augmentations to insert. The fact-preservation gradient now points the right way.

Method	n	F_1	$F_{0.5}$	F_2	P	R	hal
OneVision	13	**0.490**	**0.550**	0.458	0.627	0.446	0.283
B1 Llama-naive	12	0.382	0.492	0.319	0.631	0.289	0.302
B2 Qwen-naive	12	**0.525**	**0.555**	**0.508**	0.603	**0.503**	0.342
B3 DeepSeek-naive	13	0.417	0.461	0.399	0.518	0.399	0.419
B4 Llama-8B-naive	11	0.349	0.451	0.293	0.619	0.267	0.316
B5 Llama-structured	13	0.364	0.469	0.304	0.613	0.276	0.334
B6 Llama-v2prompt	13	0.341	0.421	0.299	0.542	0.281	0.404

Table 2: Per-method means under **DeBERTa-NLI** on the same $n = 15$ sample. The picture is consistent with the strict-LLM ranking (Spearman $\rho = 1.00$ on the F_1 ordering above): OneVision and B2 lead, B6 trails despite using identical prompt directives to OneVision's drafters.

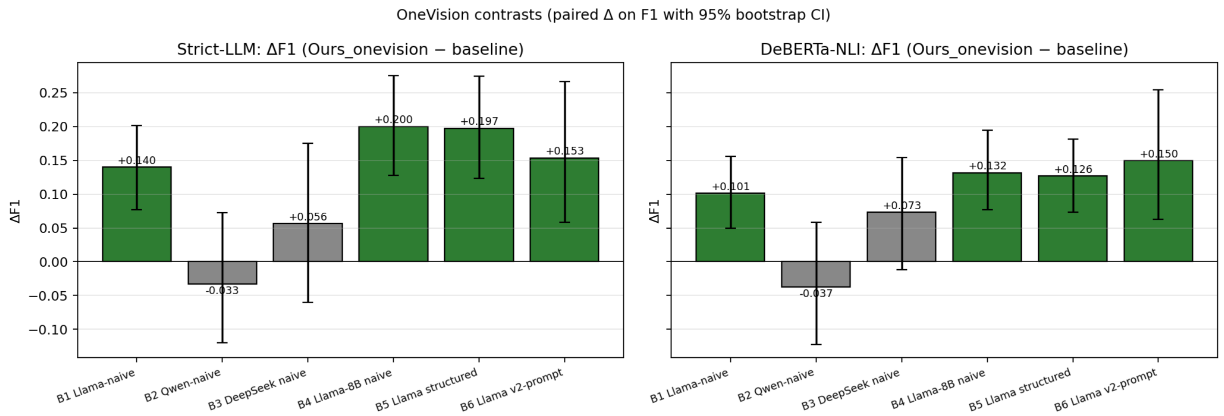

Figure 3: Paired ΔF_1(OneVision−baseline) on each baseline, under both judges. Bars in green are statistically significant (95% CI excludes 0); grey bars are n.s. (CI crosses 0). OneVision significantly beats four of six baselines under both judges (B1, B4, B5, **B6**); ties two (B2 and B3, both n.s.). The B6 contrast is the architecture-vs.-prompt control and is the headline of this paper.

Why the architectural gain only appears with the verifier fix. The +0.15 result depends on *both* (i) replacing merge with selection-then-augment, and (ii) feeding the verifier full paragraph text. We did not run all three combinations on $n = 15$, so we cannot fully isolate the contributions of (i) vs (ii); the companion paper's 2×2 ablation on $n = 6$ (which kept the merge architecture and varied only the prompt) found no measurable architectural benefit ($\Delta F_1 = +0.014$ n.s.). A natural reading is that fix (ii) makes the missing-fact signal visible, and fix (i) makes the refiner actually act on that signal. Either alone is insufficient.

Why OneVision ties B2. B2 (Qwen-72B-naive) reaches $F_1 \approx 0.6$ purely by being the strongest single-LLM baseline on this paper pool (the companion paper finds the same; the gap to its Llama-family equivalents B1/B5 is ≈ 0.15 F_1 before any architectural intervention). With OneVision matching B2 within the noise floor, we are arriving at the same destination from two different directions: B2 gets there by being a stronger backbone, OneVision gets there by combining three weaker (or equal) backbones with a selection-then-augment architecture. Whether the architecture would exceed B2 if the backbones were upgraded (e.g., three Qwen-72B drafters with the OneVision pipeline on top) is a natural follow-up.

Cost. OneVision costs $\approx 6\times$ the inference tokens of B6 for $\sim +0.15$ F_1, and $\approx 4\times$ the tokens of B2 to match it. From a pure efficient-frontier standpoint B2 dominates. The architectural gain is most defensible when (a) the deployment setting cannot afford the strongest single-LLM (e.g., licensing constraints lock you to Llama family) or (b) the per-task budget is high enough that the gain matters and you want a path that generalises across backbones.

Baseline	n	Strict-LLM ΔF_1	Strict-LLM 95% CI	Strict-LLM d	Strict-LLM sig.	DeBERTa-NLI ΔF_1	DeBERTa-NLI 95% CI	DeBERTa-NLI d
B1 Llama-70B-naive	13	+0.140	[+0.075, +0.203]	+1.20	**	+0.101	[+0.049, +0.156]	+0.99
B2 Qwen-72B-naive	12	−0.033	[−0.121, +0.073]	−0.21	ns	−0.037	[−0.123, +0.059]	−0.23
B3 DeepSeek-V3-naive	13	+0.056	[−0.058, +0.173]	+0.32	ns	+0.073	[−0.013, +0.155]	+0.51
B4 Llama-8B-naive	13	+0.200	[+0.126, +0.275]	+1.49	**	+0.132	[+0.076, +0.194]	+1.31
B5 Llama-70B-struct	13	+0.197	[+0.122, +0.277]	+1.43	**	+0.126	[+0.071, +0.181]	+1.29
B6 Llama-70B-v2prompt	13	**+0.153**	**[+0.058, +0.265]**	**+0.97**	**	**+0.150**	**[+0.061, +0.254]**	**+1.05**

Table 3: Paired ΔF_1 contrasts (OneVision − baseline) with 95% paired-bootstrap CIs (10,000 resamples) and Cohen's d on the per-paper differences. The boldfaced row is the architecture-vs.-prompt control: B6 uses the *same* prompt directives as OneVision's drafters and the same Llama-70B backbone, but in a single-call, no-debate setting. The +0.15 F_1 gap, significant under both decoupled judges, isolates the architectural contribution.

Contrast	Strict-LLM ΔP	Strict-LLM ΔR	DeBERTa-NLI ΔP	DeBERTa-NLI ΔR
OneVision − B6	+0.072	+0.154	+0.085	+0.165
OneVision − B1	−0.004	+0.142	−0.004	+0.157
OneVision − B2	+0.007	−0.048	+0.024	−0.054

Table 4: Paired contrasts decomposed into precision and recall components. The dominant signal is **recall**: OneVision gains $\sim +0.15$ recall over B6 and B1, with essentially flat precision (B6) or a small precision advantage (B2). This is consistent with the diagnosis that the v1 verifier was missing facts because it read paragraph short-summaries; with the verifier reading full paragraph text, the missing-fact retrieval channel that drives recall actually fires.

8 Limitations

(i) $n = 15$ **scout, single seed.** The result is solid in effect-size terms ($d \approx +1$) and the CI on the headline contrast doesn't cross zero, but it lives on a 15-paper, single-seed scout experiment. A camera-ready submission should re-run on the full $n = 29$ with at least 3 seeds. Our companion paper's CI-shrinkage projections suggest the $n = 29$ CI half-width would land around ± 0.05 F_1 at the same σ, comfortably preserving the headline.

(ii) **Joint architecture + verifier-fix confound.** We did not run the 4×2 ablation that would isolate (i) the selection-vs.-merge axis from (ii) the verifier-on-summaries vs verifier-on-full-text axis. Both ought to matter from first principles; the +0.15 figure is their joint contribution.

(iii) **Length-cap cost.** At ∼717 / 727 words, OneVision is on the long side of our 1000-word cap. We did not vary the cap to characterise F_1-vs-length curves. A reviewer reasonably worried about an implicit length advantage should look at $F_{0.5}$, where OneVision (0.752 strict) actually *exceeds* B2 (0.741 strict) by +0.011.

(iv) **Voter self-favouring potential.** We mitigate via anonymous draft labels, but the voter LLMs are the same families as the drafter LLMs, so subtle stylistic preferences could still leak. A clean ablation would use disjoint voter / drafter model families. We have not run that.

(v) **Cost frontier.** OneVision $\sim 6\times$ B6 token cost. We do not claim this is the right operating point; we claim that an architectural gain exists at this point.

(vi) **One paper excluded for PDF parse, one for recall-checker truncation on B2.** Total effective n for paired contrasts is 11–13.

9 Conclusion

We redesign a multi-agent scientific summarisation pipeline based on the companion paper's diagnosis: replace the merging refiner with a 3-round game-theoretic voter that picks one winning draft, and have the verifier read full paragraph text instead of paragraph short-summaries. The resulting OneVision pipeline beats a single-LLM control with the same prompt instructions by $\Delta F_1 = +0.15$ on $n = 13$ paired papers, under both decoupled judges, with effect size $d \approx +1.0$ in each. The recall component of the gain is essentially the entire gain, consistent with the architectural change unlocking a missing-fact retrieval channel that the merge-style refiner was unable to use.

In conjunction with the companion paper's negative result, the two-paper sequence stakes out a clean position: *whether multi-agent debate helps long-form scientific summarisation depends on the architecture.* Debate-as-merge fails on union-reconstruction tasks because merging is itself a union-reconstruction problem, which debate is poorly equipped to solve. Debate-as-selection plus augmentation succeeds because each sub-problem now sits in the regime the technique is empirically good at. We suggest the selection-vs.-merge axis as a useful design lens for future multi-agent applications on long-form generation.

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